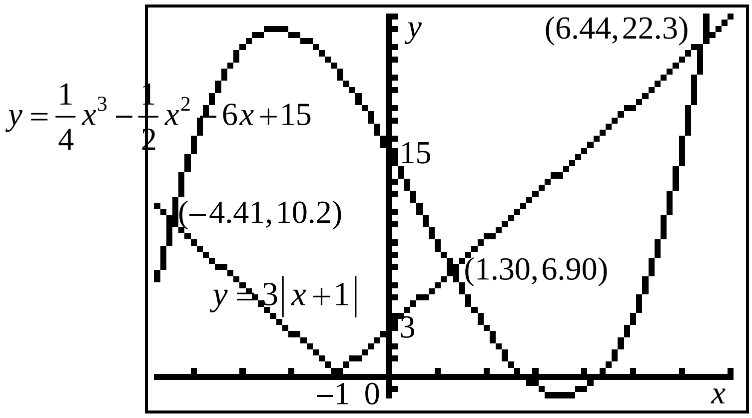
2009 RI(JC) H2 Maths Preliminary Examination Paper 1 (Solutions)

1



From the graphic calculator, the x-coordinate of the 3 points of intersection are: 4.4140, 1.3005 and 6.4352. (5 s.f.)

Now, $|x+1| \le \frac{1}{12}x^3 - \frac{1}{6}x^2 - 2x + 5$

$\Rightarrow 3|x+1| \le \frac{1}{4}x^3 - \frac{1}{2}x^2 - 6x + 15$.

Hence, from the graph, $-4.41 \le x \le 1.30$ or $x \ge 6.44$. (3 s.f.) □

2

$y^2 + e^x + 3y = 5 ----------(1)$

(i) Differentiating (1) with respect to x,

$$2y\frac{dy}{dx} + e^x + 3\frac{dy}{dx} = 0$$

$$(2y+3)\frac{dy}{dx} + e^x = 0 \quad --------(2)$$

Differentiating (2) with respect to x,

$$\left(2\frac{dy}{dx}\right)\left(\frac{dy}{dx}\right) + (2y+3)\frac{d^2y}{dx^2} + e^x = 0$$

$$2\left(\frac{dy}{dx}\right)^2 + (2y+3)\frac{d^2y}{dx^2} + e^x = 0 \text{ (Shown)} \quad ------(3)$$

(ii) $y^2 + e^x + 3y = 5 ----------(1)$

When $x = 0$, $y^2 + 3y - 4 = 0$ (from (1))

$(y-1)(y+4) = 0$

$\Rightarrow y = 1$ or -4 (NA since $y > 0$)

When $x = 0$, $\frac{dy}{dx} = -\frac{1}{5}$ (from (2))

$\frac{d^2y}{dx^2} = -\frac{27}{125}$ (from (3))

So $y = 1 - \frac{1}{5}x - \frac{27}{250}x^2 + \ldots$

3	Floor area = $4+\left(\frac{1}{2}\right)(2)(2)\sin\theta$ $=4+2\sin\theta$ Unit cost of carpeting $=\dfrac{8000}{4+2\sin\theta}$ $\approx\dfrac{8000}{4+2\theta}$ $=4000(2+\theta)^{-1}$ $=4000\left(\frac{1}{2}\left(1+\frac{\theta}{2}\right)^{-1}\right)$ $\approx 2000\left(1-\frac{\theta}{2}\right)$ $=\$(2000-1000\theta)$ /m^2 So a = 2000, b = –1000
4(i) **(ii)**	$\int\sec^3 x\tan x\ \mathrm{d}x=\int(\sec x\tan x)(\sec^2 x)\ \mathrm{d}x$ $=\frac{1}{3}\sec^3 x+c$ Let $u=\sqrt{x}$. Then $\dfrac{\mathrm{d}x}{\mathrm{d}u}=2u$ $\int\dfrac{1}{2\sqrt{x}(x-9)}\mathrm{d}x$ $=\int\dfrac{1}{2u(u^2-9)}(2u)\ \mathrm{d}u$ (I) $=\int\dfrac{1}{u^2-9}\mathrm{d}u$ $=\frac{1}{6}\ln\left\|\dfrac{u-3}{u+3}\right\|+c$ $=\frac{1}{6}\ln\left\|\dfrac{\sqrt{x}-3}{\sqrt{x}+3}\right\|+\mathrm{c}$
5(i)	Since $2(2)-0-2=2$, the point with position vector $2\mathbf{i}+2\mathbf{k}$ lies on Π_2. A direction vector for the line of intersection is $\begin{pmatrix}1\\-3\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\-1\end{pmatrix}=\begin{pmatrix}5\\5\\5\end{pmatrix}$. An equation of line of intersection, ℓ, is $\mathbf{r}=\begin{pmatrix}2\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\1\end{pmatrix},\ \lambda\in\mathbb{R}$

5(ii)

From GC

$$\begin{cases} x-3y+2z=6 \\ 2x-y-z=2 \\ x-3y-z=-12 \end{cases}$$

$\Rightarrow x=6, y=4, z=6$

The 3 planes intersect at a common point with position vector $\begin{pmatrix}6\\4\\6\end{pmatrix}$.

5(iii)

For the 3 planes to intersect at a common line, line ℓ must lie on plane Π_3.

$$\left[\begin{pmatrix}2\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\1\end{pmatrix}\right]\cdot\begin{pmatrix}1\\b\\-1\end{pmatrix}=4b \quad \text{--- (*)}$$

$(2+\lambda)+b\lambda-(2+\lambda)=4b$

$b\lambda=4b$

$\lambda=4$ or $b=0$

Equation (*) must be true for all real values of λ.

So $b = 0$.

<u>Alternative Method 1</u>

$$\begin{pmatrix}2\\0\\2\end{pmatrix}\cdot\begin{pmatrix}1\\b\\-1\end{pmatrix}=4b \Rightarrow b=0 \text{ and } \begin{pmatrix}3\\1\\3\end{pmatrix}\cdot\begin{pmatrix}1\\b\\-1\end{pmatrix}=4b \Rightarrow b=4b \Rightarrow b=0$$

Since 2 points on the line ℓ lies on the plane Π_3, the line ℓ must then lie on the plane Π_3. Thus $b = 0$.

<u>Alternative Method 2</u>

$$\begin{pmatrix}2\\0\\2\end{pmatrix}\cdot\begin{pmatrix}1\\b\\-1\end{pmatrix}=4b \Rightarrow b=0 \text{ and}$$

$$\begin{pmatrix}1\\1\\1\end{pmatrix}\cdot\begin{pmatrix}1\\b\\-1\end{pmatrix}=0 \Rightarrow b=0$$

Since point P on line ℓ lies on plane Π_3 and line ℓ is parallel to Π_3, therefore line ℓ must then lie on the plane Π_3. Thus $b = 0$.

<u>Alternative Method 3</u>

If the 3 planes intersect at a common line, the common line of intersection between any of the 2 planes must be the same.

For the line of intersection between π_1 and π_2 to be the same as the line of intersection between π_1 and π_3, P must lie on π_3 and the 2 lines must be parallel.

	$\begin{pmatrix}2\\0\\2\end{pmatrix} \cdot \begin{pmatrix}1\\b\\-1\end{pmatrix} = 4b \Rightarrow b = 0$ and $\begin{pmatrix}1\\-3\\2\end{pmatrix} \times \begin{pmatrix}1\\b\\-1\end{pmatrix} = \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$ for some λ $\Rightarrow \begin{pmatrix}3-2b\\3\\b+3\end{pmatrix} = \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\Rightarrow \lambda = 3$ and $3-2b=3$ and $b+3=3$ $\Rightarrow b = 0$ Thus $b = 0$.
6(i)	Since $\overrightarrow{MN} = \begin{pmatrix}3\\\frac{3}{2}\\\frac{5}{2}\end{pmatrix}$ and $\overrightarrow{BH} = \begin{pmatrix}-4\\3\\5\end{pmatrix}$ the acute angle between the lines l_1 and l_2 $= \cos^{-1}\left(\left\|\dfrac{\begin{pmatrix}3\\\frac{3}{2}\\\frac{5}{2}\end{pmatrix} \bullet \begin{pmatrix}-4\\3\\5\end{pmatrix}}{\left\|\begin{pmatrix}3\\\frac{3}{2}\\\frac{5}{2}\end{pmatrix}\right\| \left\|\begin{pmatrix}-4\\3\\5\end{pmatrix}\right\|}\right\|\right) = \cos^{-1}\left(\left\|\dfrac{-12+\frac{9}{2}+\frac{25}{2}}{\sqrt{50}\sqrt{17.5}}\right\|\right) = 80.3°$
6(ii)	A vector normal to the plane $EFGH$ is $\mathbf{k}$ The acute angle between the plane $EFGH$ and a line parallel to $\overrightarrow{AP}$ $= \sin^{-1}\left(\left\|\dfrac{\frac{1}{10}\begin{pmatrix}12\\21\\35\end{pmatrix} \bullet \begin{pmatrix}0\\0\\1\end{pmatrix}}{\frac{1}{100}\left\|\begin{pmatrix}12\\21\\35\end{pmatrix}\right\| \left\|\begin{pmatrix}0\\0\\1\end{pmatrix}\right\|}\right\|\right) = \sin^{-1}\left(\left\|\dfrac{3.5}{\sqrt{18.1}}\right\|\right) = 55.4°$
7(i)	For the population to grow, $u_{n+1} - u_n$ $= u_n - qu_n - 4$ > 0 Re-arranging, we get $u_n > \dfrac{4}{1-q}$.
7(ii)	Given that $u_1 = 10$ and $u_2 = 15$, we have $15 = (2-q)(10) - 4$ which gives $q = 0.1$.

The recurrence relation is then $u_{n+1} = 1.9u_n - 4$.

Using the GC to evaluate the terms, we have $u_{20} = 1000000$ (1 s.f.)

This is highly impractical as the formulation of the recurrence relation did not take into account the physical constraints of the jar.

7(iii) The suggestion should not be taken up as the differential equation gives a continuous model as an approximation which only works when the population is large enough.

8

Let P_n be the statement

$$\sum_{r=2}^{n} r^2\left[(r-1)!\right] = (n+1)! - 2 \text{ where } n \in \mathbb{Z}^+,\ n \geq 2.$$

When $n = 2$,

LHS $= 2^2(1!) = 4$, RHS $= 3! - 2 = 6 - 2 = 4$

Hence, P_2 is true.

Assume P_k is true for some $k \in \mathbb{Z}^+$, $k \geq 2$.

i.e. $\displaystyle\sum_{r=2}^{k} r^2\left[(r-1)!\right] = (k+1)! - 2$.

Want to prove that P_{k+1} is true,

i.e. to prove $\displaystyle\sum_{r=2}^{k+1} r^2\left[(r-1)!\right] = (k+2)! - 2$.

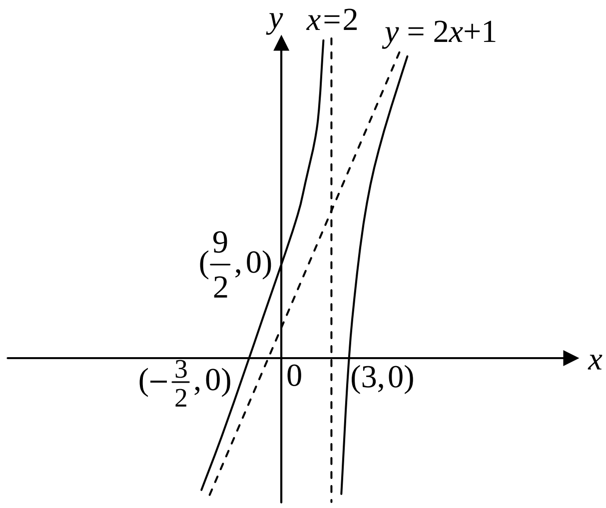
$$\begin{aligned}
\text{LHS} &= \sum_{r=2}^{k} r^2\left[(r-1)!\right] + (k+1)^2(k!) \\
&= (k+1)! - 2 + (k+1)\left[(k+1)!\right] \\
&= (k+1)!(1+k+1) - 2 \\
&= (k+2)! - 2 \\
&= \text{RHS}
\end{aligned}$$

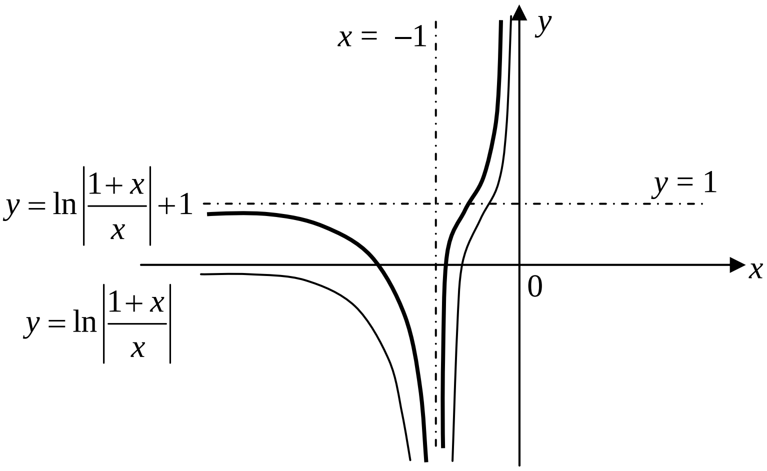
Hence P_{k+1} is true whenever P_k is true. Since P_2 is true, by using mathematical induction, P_n is true for all $n \in \mathbb{Z}^+$, $n \geq 2$.

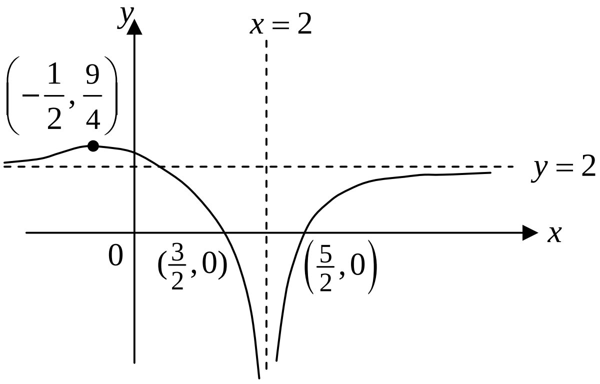
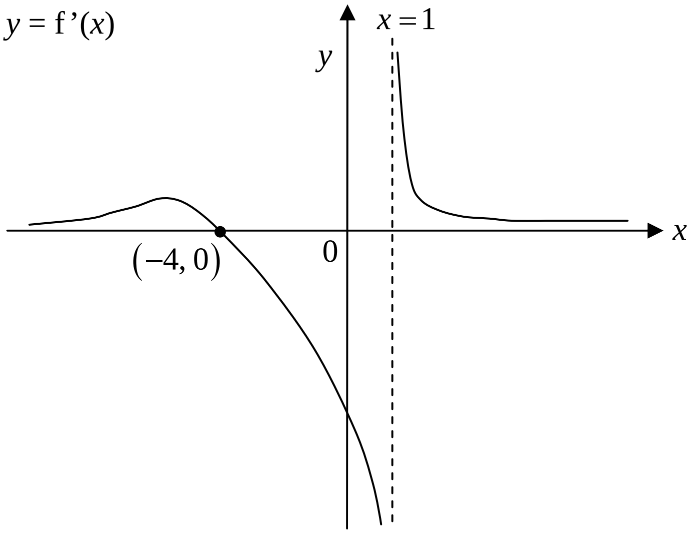
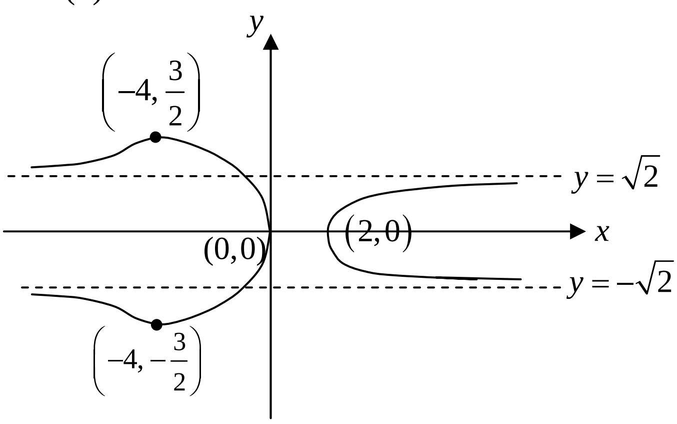
Replacing $(r+1)$ with r

$$\begin{aligned}
\sum_{r=2}^{n-1} (r+1)^2(r!) &= \sum_{r=3}^{n} r^2\left[(r-1)!\right] \\
&= \sum_{r=2}^{n} r^2\left[(r-1)!\right] - 2^2(2-1)! \\
&= (n+1)! - 2 - 4 \\
&= (n+1)! - 6
\end{aligned}$$

	Alternatively Replacing r with $(r+1)$ $$\sum_{r=2}^{n} r^2\left[(r-1)!\right] = \sum_{r=1}^{n-1}(r+1)^2\left(r!\right)$$ $$= \sum_{r=2}^{n-1}(r+1)^2\left(r!\right) + 2^2(1)!$$ $$\sum_{r=2}^{n-1}(r+1)^2\left(r!\right) = \sum_{r=2}^{n} r^2\left[(r-1)!\right] - 2$$ $$= (n+1)! - 4 \quad -2$$ $$= (n+1)! - 6$$
9	Equating the two equations, we have $$x^4 + x^2 - \frac{3}{4} = 0$$ Solving, $\quad x = \frac{1}{\sqrt{2}}$ or $x = -\frac{1}{\sqrt{2}}$ $$y = \frac{1}{2} \quad \text{or} \quad y = \frac{1}{2}$$ The coordinates of point A and B are $(-\frac{1}{\sqrt{2}}, \frac{1}{2})$ and $(\frac{1}{\sqrt{2}}, \frac{1}{2})$ respectively.
(i)	Area of R = $2\left(\int_0^{\frac{1}{\sqrt{2}}} x^2 \, dx + \int_{\frac{1}{\sqrt{2}}}^{\frac{\sqrt{3}}{2}} \sqrt{\frac{3}{4} - x^2} \, dx\right)$ = 2 (0.11785 + 0.05403) = 0.34 (2 d.p.)
(ii)	Volume $= \pi\int_0^{\frac{1}{2}} \left(\frac{3}{4} - y^2 - y\right) dy$ $$= \pi\left[\frac{3}{4}y - \frac{y^3}{3} - \frac{y^2}{2}\right]_0^{\frac{1}{2}}$$ $$= \frac{5}{24}\pi$$
10(i)	$b = -2$
10(ii)	$$y = \frac{2x^2 + ax + 3a}{x + b}$$ $$\frac{dy}{dx} = \frac{(4x+a)(x-2) - (2x^2 + ax + 3a)}{(x-2)^2} = \frac{(2x^2 - 8x - 5a)}{(x-2)^2}$$ For C to have no stationary points, $\frac{dy}{dx} = 0$ has no real roots. i.e. $2x^2 - 8x - 5a = 0$ has no real roots $\Rightarrow$ Discriminant = $64 - 4(2)(-5a) < 0$ $\Rightarrow \; 8 + 5a < 0$, so $a < -\frac{8}{5}$ (Ans)

	Alternative Method For C to have no stationary points, $\frac{\mathrm{d}y}{\mathrm{d}x} \neq 0$ i.e. $2x^2 - 8x - 5a \neq 0$ $2(x-2)^2 - (8+5a) \neq 0$ Since $2(x-2)^2 \geq 0$ for all $x \in \mathbf{R}$, $\Rightarrow 8+5a < 0 \Rightarrow a < -\frac{8}{5}$ (Ans)
10(iii)	Given that $a = -3$, $y = \frac{2x^2 - 3x - 9}{x-2} = 2x + 1 - \frac{7}{x-2}$ 
10(iv)	Scale C parallel to the y-axis (with x-axis invariant) by a factor of 1/3, then translate the resulting curve by 5 units in the positive y-direction to obtain the required curve. **Alternatively,** Translate C 15 units in the positive y-direction, then scale parallel to the y-axis (with x-axis invariant) by a factor of 1/3 to obtain the required curve.
11(i)	$\frac{\mathrm{d}y}{\mathrm{d}x} \propto \frac{1}{x(1+x)} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k}{x(1+x)}$ for some real constant k. Since $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2}$ when $x = -2$, $-\frac{1}{2} = \frac{k}{-2(-1)} \Rightarrow k = -1$. $\therefore \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{x(1+x)}$ (Shown) $\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{1+x} - \frac{1}{x}$ $\Rightarrow y = \int \left(\frac{1}{1+x} - \frac{1}{x} \right) \mathrm{d}x = \ln\lvert 1+x \rvert - \ln\lvert x \rvert + C$ or $y = \ln\left\lvert \frac{1+x}{x} \right\rvert + C$
11(ii)	$y = \ln 3$ when $x = -\frac{1}{4}$ $\Rightarrow \ln 3 = \ln\left\lvert \frac{1-{}^1\!/_4}{-{}^1\!/_4} \right\rvert + C = \ln\lvert -3 \rvert + C \Rightarrow C = 0$ $\therefore y = \ln\lvert 1+x \rvert - \ln\lvert x \rvert$ or $y = \ln\left\lvert \frac{1+x}{x} \right\rvert$

11(iii)	
12(a) (i)	First term of first row $=1$ First term of second row $=1+1$ First term of third row $=1+1+2$ First term of fourth row $=1+1+2+3$ First term of n^{th} row $=1+1+2+3+......+(n-1)$ $=1+\frac{n-1}{2}(1+(n-1))$ $=1+\frac{1}{2}n(n-1)$
(ii)	Sum of the sum of all the elements n^{th} row $=\frac{n}{2}(2a+d(n-1))$ $=\frac{n}{2}\left(2\left(1+\frac{1}{2}n(n-1)\right)+1(n-1)\right)$ $=\frac{1}{2}n(n^2+1)$
	For $\frac{1}{2}n(n^2+1)>10^5$ When $n=58,$ sum of elements $=97585$ When $n=59,$ sum of elements $=102719$ Least value of n is 59.
12(b)	$S_4=\frac{16}{17}S_8$ $\frac{a(1-r^4)}{1-r}=\frac{16}{17}\frac{a(1-r^8)}{1-r}$ $17-17r^4=16-16r^8$ $16r^8-17r^4+1=0$ From GC, $r=\frac{1}{2}$ Sum to infinity $=\frac{a}{1-\frac{1}{2}}=2a$.

13(i)	$y = \mathrm{f}(2x-3)$ 
13(ii)	$y = \mathrm{f}'(x)$ 
13(iii)	$y^2 = \mathrm{f}(x)$ 
14(i) (a)	$zz^* + \mathrm{i}a(z-z^*) + 2a^2 = 2 + a(z+z^*),\ a \in \mathbb{R}$ $x^2 + y^2 + ia\ (z-z^*) - a\ (z+z^*) + 2a^2 = 2$ $x^2 + y^2 + ia\ (2iy) - a\ (2x) + 2a^2 = 2$ $x^2 - 2ax + a^2 + y^2 - 2ay + a^2 = 2$ $(x-a)^2 + (y-a)^2 = 2$ (shown)
(b)	The equation $(x-1)^2 + (y-1)^2 = 2$ is equivalent to $\lvert z-(1+\mathrm{i})\rvert = \sqrt{2}$. Since $\lvert 1+i \rvert = \sqrt{2}$ and $\arg(1+i) = \dfrac{\pi}{4}$, then z and 1 + i both lie on a straight line

	passing through the origin and z is at a distance of $\sqrt{2}$ away from 1 + i. It is then clear to see that $\lvert z\rvert = 2\sqrt{2}$, so $z^* = 2\sqrt{2}\mathrm{e}^{-\frac{\pi i}{4}}$. Alternatively, students may show the above in an Argand diagram. $(z^*)^n = \left(2\sqrt{2}\mathrm{e}^{-\frac{\pi i}{4}}\right)^n$ is purely imaginary $\Rightarrow\ \mathrm{e}^{-\frac{n\pi \mathrm{i}}{4}} = \mathrm{e}^{-(2m+1)\frac{\pi}{2}\mathrm{i}}$ where $m \in \mathbf{Z},\ m \geq 0$ $\Rightarrow\ \frac{n}{4} = \frac{(2m+1)}{2}$ $\Rightarrow\ n = 4m+2$ where $m \in \mathbf{Z},\ m \geq 0$. **Alternatively**, $(z^*)^n = \left(2\sqrt{2}\mathrm{e}^{-\frac{\pi i}{4}}\right)^n$ is purely imaginary $\Rightarrow \mathrm{Re}\left(z^*\right)^n = 0 \Rightarrow \cos\left(\frac{-n\pi}{4}\right) = 0 \Rightarrow \frac{-n\pi}{4} = -(2m+1)\frac{\pi}{2}$ $\Rightarrow\ n = 4m+2$ where $m \in \mathbf{Z},\ m \geq 0$.
14(ii)	$\left(w-1-i\right)^5 = 4\sqrt{2}$ $\left(w-1-i\right)^5 = 4\sqrt{2}\mathrm{e}^{2k\pi i}$ $\left(w-1-i\right) = \sqrt{2}\mathrm{e}^{\frac{2k\pi i}{5}},\ k = 0,\ \pm 1,\ \pm 2$ $w = 1+i+\sqrt{2}\mathrm{e}^{\frac{2k\pi i}{5}},\ k = 0,\ \pm 1,\ \pm 2$ The points lie on the circle $(x-1)^2+(y-1)^2 = 2$

END